

**AI-BASED INVESTMENT PLAN  
RECOMMENDATION MODULE  
FOR DIGITAL WELFARE  
MANAGEMENT IN SRI LANKA**

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## DECLARATION OF THE CANDIDATE AND SUPERVISOR

I declare that this is my own work and this report does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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(Ms. Geethanjali Wimalaratne)

## ABSTRACT

This research focuses on the design and implementation of an AI-Based Investment Plan Recommendation Module aimed at empowering low-income welfare recipients in Sri Lanka. By leveraging machine learning algorithms, the system analyzes user income, expense patterns and financial risk profiles to recommend safe, practical and sustainable investment or savings opportunities. The module is part of a larger AI-powered Digital Welfare Platform, ensuring that welfare beneficiaries receive not only immediate financial aid but also guidance for long-term economic stability.

The proposed solution combines clustering techniques to segment users by financial capacity and collaborative filtering to match them with investment schemes aligned to their needs. The methodology includes data collection and preprocessing, implementation of K-Means clustering algorithms to segment welfare recipients into distinct financial groups, development of a collaborative filtering-based recommendation engine and integration with citizen and officer dashboards. The system incorporates progress tracking mechanisms to monitor user engagement with recommended plans and adaptive follow-up scheduling to provide timely interventions based on tracked progress.

Recommendations are personalized, transparent and compliant with ethical AI principles. The system addresses critical research gaps including the lack of personalized financial advice for welfare beneficiaries and the absence of continuous monitoring mechanisms. Experimental results demonstrate that the clustering algorithm effectively segments users into meaningful financial groups and the recommendation engine provides accurate, contextually relevant investment suggestions. Ultimately, this module aims to reduce dependency on welfare, promote financial literacy and improve recipients' ability to participate in the economy.

**Keywords:** Financial AI, Recommender Systems, Clustering, Low-Income Empowerment, Digital Welfare Systems, K-Means Clustering, Collaborative Filtering

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## LIST OF ABBREVIATIONS

|             |                                   |
|-------------|-----------------------------------|
| <b>AI</b>   | Artificial Intelligence           |
| <b>API</b>  | Application Programming Interface |
| <b>AWS</b>  | Amazon Web Services               |
| <b>CSV</b>  | Comma-Separated Values            |
| <b>GIS</b>  | Geographic Information System     |
| <b>HTML</b> | HyperText Markup Language         |
| <b>HTTP</b> | HyperText Transfer Protocol       |
| <b>IFC</b>  | International Finance Corporation |
| <b>IT</b>   | Information Technology            |
| <b>JSON</b> | JavaScript Object Notation        |
| <b>ML</b>   | Machine Learning                  |
| <b>NGO</b>  | Non-Governmental Organization     |
| <b>NLP</b>  | Natural Language Processing       |
| <b>PDPA</b> | Personal Data Protection Act      |
| <b>REST</b> | Representational State Transfer   |
| <b>SaaS</b> | Software as a Service             |
| <b>SQL</b>  | Structured Query Language         |
| <b>UI</b>   | User Interface                    |
| <b>UX</b>   | User Experience                   |

## 1. INTRODUCTION

This research project aims to design and develop an AI-powered Investment Plan Recommendation Module as part of a broader digital welfare management platform for Sri Lanka. The country's social protection system offers multiple programs, including Samurdhi, disability allowances, elderly pensions and nutritional aid. While these schemes address immediate needs, they often lack mechanisms to guide beneficiaries toward long-term financial independence. Many recipients remain dependent on welfare due to limited financial literacy, lack of tailored guidance and absence of accessible tools to evaluate safe investment opportunities.



*Figure 1: Current Sri Lanka Welfare Portal*

Current welfare administration faces challenges such as fragmented data handling, absence of personalized financial advisory services and lack of predictive systems for economic empowerment. Internationally, countries have integrated AI-driven financial planning tools into welfare programs to promote self-sufficiency. For example, micro-investment advisory systems in India and Bangladesh have demonstrated significant improvements in household savings and income diversification. However, Sri Lanka has yet to adopt a unified, AI-enabled financial recommendation solution within its welfare ecosystem.

The proposed module is designed to enhance not replace existing welfare processes by offering personalized, risk-aware investment guidance to low-income beneficiaries. It uses machine learning algorithms to analyze user profiles, including income, expenses, savings patterns and risk tolerance, to recommend suitable micro-investment and savings plans. The system will incorporate clustering techniques to segment users by financial capacity and collaborative filtering methods to match them with relevant opportunities from government, NGOs and regulated financial institutions.

Functioning as part of the integrated welfare platform, this module will benefit three key user groups:

- Citizens: Gain accessible, personalized financial recommendations in their preferred language.
- Officers: Monitor adoption and outcomes of financial plans for welfare recipients.
- Policymakers: Access analytics to evaluate program impact and optimize resource allocation.

By embedding AI-driven investment guidance into the welfare system, this research aims to reduce long-term dependency on aid, promote financial stability, and contribute to sustainable socio-economic development in Sri Lanka. The module integrates with the overall digital welfare platform, which includes eligibility prediction engines, healthcare recommendation systems, multilingual chatbot support and AI-GIS welfare insights, creating a comprehensive ecosystem for welfare service delivery.

### **1.1 Background and Literature Survey**

Financial inclusion and effective financial guidance have become critical components in addressing economic disparities faced by low-income populations worldwide. According to the World Bank's Global Findex Database, about 1.7 billion adults remain unbanked globally, lacking access to basic financial services. This exclusion often results in reduced opportunities for wealth accumulation, limited access to credit and increased vulnerability to economic shocks. In Sri Lanka, a significant portion of welfare recipients experience similar barriers, as the existing financial systems and advisory services do not sufficiently address their specific needs and constraints.

Traditional financial advisory services are typically costly and inaccessible to low-income groups, who require affordable, personalized support to make informed decisions about investments and savings. The advent of digital financial services powered by artificial intelligence (AI) offers a promising solution to bridge this gap. AI-driven financial guidance platforms can democratize access by providing micro-investment recommendations and financial planning tailored to each user's socio-economic background, risk appetite and income dynamics. These systems leverage data such as income patterns, expense tracking and eligibility for various government aid programs to deliver individualized and scalable advice.

Key benefits of AI-powered financial guidance systems include:

- Improved Accessibility: These platforms make financial advice available to users in remote or underserved regions via mobile or web applications, bypassing the need for physical financial advisors.
- Cost Reduction: Automation lowers service delivery costs, enabling sustainable support for economically disadvantaged groups.

- **Personalized Recommendations:** Advanced clustering algorithms segment users into homogeneous groups, enabling more relevant investment options and financial aid schemes tailored to specific profiles.
- **Dynamic Adaptation:** Recommender systems constantly update advice based on changing financial situations, allowing users to adjust plans as their circumstances evolve.

Despite these advantages, implementing AI in welfare-focused financial guidance raises several challenges:

- **Data Sensitivity and Privacy:** Handling financial and personal data of vulnerable populations requires stringent security and ethical frameworks to protect user confidentiality.
- **User Digital Literacy:** Many low-income individuals have limited experience with digital technologies; thus, systems must be intuitive and provide trustworthy human-centered support.
- **Algorithmic Fairness:** AI models must be carefully developed and monitored to avoid perpetuating biases or excluding certain user groups due to biased training data or flawed assumptions.
- **Integration with Welfare Programs:** Coordination between AI financial guidance and existing welfare schemes is often lacking, reducing effectiveness and uptake among beneficiaries.

A review of existing literature reveals growing evidence supporting the role of AI in enhancing financial inclusion:

- The International Finance Corporation (IFC) reported in 2023 that AI-enabled micro-investment platforms increased savings rates by up to 35% in underserved communities across Asia and Africa, demonstrating substantial economic empowerment effects.
- Chen et al. (2022) illustrated that clustering algorithms effectively segmented low-income populations for credit risk scoring and personalized investment suggestions, leading to improved financial decision-making accuracy.
- Kumar and Singh (2021) found that recommender systems enhanced user engagement and satisfaction when transparent explanations of financial advice were incorporated, building critical trust in AI tools for low-income users.

Nonetheless, few platforms explicitly focus on integrating welfare program data and addressing the multi-dimensional needs of welfare recipients. The proposed AI-based financial guidance module uniquely aims to fill this gap within the Sri Lankan social protection ecosystem, supporting users with tailored micro-investment plans and financial aid recommendations that amplify the benefits of digital welfare services.

By synthesizing insights from recent advances in AI, behavioral economics, and social welfare policy, this research sets out to develop a financial guidance system that is not only technically robust but socially inclusive and contextually relevant. The goal is to empower vulnerable citizens with smart, evidence-based financial advice, thus fostering enhanced economic resilience, financial literacy, and improved welfare outcomes.

## **1.2 Research Gap**

A research gap refers to an area within a field of study that has not been adequately explored or addressed by prior studies. In our investigation related to financial guidance for low-income welfare recipients, several key gaps have been identified.

The first significant research gap concerns the lack of personalized, evidence-based financial advice tailored to the unique needs of welfare beneficiaries. Current financial aid programs often allow recipients to self-select financial support options or micro-investment plans based on limited information or personal preference. However, this approach carries risks, as individuals might choose options that seem beneficial but may ultimately worsen their financial stability. For instance, a beneficiary may opt for a high-risk micro-investment appealing in the short term, but if market conditions turn unfavorable, this could increase their financial vulnerability instead of reducing it. This gap underlines the need for intelligent recommendation systems that factor in users' income, expense patterns, and risk profiles to suggest optimized and safer financial plans.

The second gap is the absence of continuous monitoring and feedback mechanisms that track the financial progress and investment outcomes of welfare recipients after advice delivery. Existing systems rarely provide ongoing support or detailed insights into how users' financial behaviors evolve over time. Consequently, advisors and policymakers lack real-time understanding of beneficiaries' financial health post-intervention, limiting their ability to offer timely adjustments or additional guidance. Without such monitoring, the system cannot dynamically update financial recommendations or identify recipients who may be struggling, leading to inefficient use of welfare resources and missed opportunities for economic empowerment.

Addressing these research gaps by integrating AI-powered clustering techniques and recommender systems into a financial guidance module can improve personalization, risk management, and adaptive feedback. This innovation is crucial to enhancing the welfare ecosystem and ensuring low-income citizens receive the most effective and sustainable financial support.

*Table 1.1: Comparison of Key Features Between Proposed AI-Powered Financial Module  
and Existing Advisory Methods*

| <b>Feature</b>                                | <b>Proposed Platform</b> | <b>Current Sri Lanka Systems</b> |
|-----------------------------------------------|--------------------------|----------------------------------|
| Personalized Micro-Investment Recommendations | Yes                      | No or Limited                    |
| Clustering Based on Financial Behavior        | Yes                      | No                               |
| Continuous Progress Tracking & Monitoring     | Yes                      | No                               |
| Adaptive Follow-up Scheduling                 | Yes                      | No                               |
| Integration with Welfare Program Data         | Yes                      | No                               |
| Multilingual Support                          | Yes                      | Partial                          |

### **1.3 Research Problem**

The research problem defines the gap in existing knowledge that this study seeks to address. Below are the primary research questions identified concerning the AI-based financial guidance module:

- How to ensure that the recommended financial plans or micro-investment options effectively improve the financial well-being of low-income welfare recipients?

The first challenge is to identify and validate financial advice or investment recommendations that genuinely benefit beneficiaries without exposing them to undue risk. The approach involves conducting a thorough review of financial inclusion literature and consulting financial experts specializing in low-income populations. Additionally, engaging with welfare recipients to understand their preferences and financial behaviors is crucial. This ensures that recommendations are both effective and acceptable to users, thereby maximizing real-world applicability. The ultimate goal is to design recommendation algorithms that consistently suggest financial options proven to enhance economic stability for welfare users.

- How to create reliable methods to monitor and track financial behavior and outcomes of users after recommendations are given?

The second research problem focuses on developing mechanisms to accurately track beneficiaries' adherence to the recommended plans and monitor their financial progress over time. This involves designing data collection methods and analytic tools to assess

measurable impacts, such as saving rates, investment engagement, or improvements in financial literacy. Ensuring the tracking system is effective and user-friendly is critical for maintaining continuous support and enabling timely interventions if users deviate from advised actions.

- How to accurately predict and adapt follow-up financial guidance or scheduling based on users' progress?

The third research problem is to create a dynamic, responsive mechanism that adjusts the frequency and type of financial advice or support sessions according to the tracked progress of beneficiaries. Rather than relying on fixed schedules, the system will analyze ongoing user data to predict when a user requires additional engagement or re-evaluation of guidance. For example, if a user is demonstrating consistent progress and adherence, less frequent follow-ups might be scheduled. Conversely, if a user is struggling with adherence or facing financial setbacks, the system recommends earlier or intensified support to prevent worsening financial vulnerability.

## **1.4 Research Objectives**

### **1.4.1 Main Objective**

The main objective of this research is to enhance the efficiency and effectiveness of AI-driven financial guidance for low-income welfare recipients in Sri Lanka. This innovation aims to empower beneficiaries by providing personalized financial advice, micro-investment recommendations, and dynamic support based on their financial behaviors and profiles. To deliver a more tailored and adaptive financial empowerment experience, the research emphasizes the design and integration of intelligent clustering algorithms, recommender systems, progress tracking mechanisms, and adaptive follow-up scheduling within a unified digital welfare platform.

To accomplish the above main objective, the following four sub-objectives must be achieved:

- Develop clustering algorithms to segment welfare recipients based on financial backgrounds, income patterns, and risk profiles to facilitate precise targeting of financial recommendations.
- Implement a financial recommendation system that suggests suitable micro-investment plans and aid options customized to individual beneficiary needs and preferences.



- Create a mechanism to monitor and analyze user engagement with the recommended financial plans over time, capturing adherence, usage patterns, and financial outcomes.
- Design an adaptive scheduling module to recommend timely follow-up advisory sessions or interventions based on the tracked progress and changing financial conditions of users.

#### **1.4.2 Specific Objectives**

The core component for which I am responsible involves delivering an AI-driven financial guidance module that recommends appropriate micro-investment and financial aid plans, monitors beneficiary engagement, and schedules subsequent advisory interactions. To achieve this, I have further subdivided my work into the following specific sub-objectives:

- Provide personalized financial plans by using AI to curate tailored, realistic investment and aid options aligned with each low-income user's financial situation and goals.
- Track beneficiary adherence and progress by monitoring their saving habits, investment participation, and aid use to evaluate the guidance's effectiveness.
- Deliver actionable insights to officers via diagnostic reports that provide visibility into user behavior and improvement, informing interventions and policy adjustments.
- Predict and schedule follow-ups by using progress data to forecast when a beneficiary needs guidance, prioritizing users in distress over stable ones.

## **2. METHODOLOGY**

### **2.1 Understanding Key Pillars of the Research Domain**

#### **2.1.1 Machine Learning and Clustering**

Machine learning (ML) is a subset of artificial intelligence that enables systems to learn and improve from experience without being explicitly programmed. In the context of this research, machine learning provides the foundation for developing intelligent systems that can analyze financial data, identify patterns, and make predictions about user behavior and needs.

Clustering is an unsupervised machine learning technique used to group similar data points together based on their characteristics. For this project, clustering algorithms segment welfare recipients into distinct groups based on their financial profiles, including income levels, expense patterns, family size, employment status and risk tolerance. This segmentation enables the system to provide more targeted and relevant financial recommendations to each group.

The K-Means clustering algorithm is particularly well-suited for this application because it efficiently partitions large datasets into a specified number of clusters by minimizing the within-cluster variance. The algorithm iteratively assigns data points to the nearest cluster center and updates these centers until convergence is reached. This approach allows the system to identify natural groupings in the welfare recipient population, such as low-risk savers, moderate-risk investors and high-need beneficiaries requiring immediate assistance.

#### **2.1.2 Collaborative Filtering and Recommendation Systems**

Collaborative filtering is a widely-used technique in recommendation systems that predicts user preferences based on the preferences and behaviors of similar users. In financial guidance systems, collaborative filtering analyzes historical data about which financial products or investment plans have been successful for users with similar profiles, and then recommends those products to new users with comparable characteristics.

The recommendation engine developed in this research combines collaborative filtering with content-based filtering approaches. Content-based filtering considers the attributes of financial products themselves, such as risk level, return potential, liquidity, and minimum investment amount. By combining both approaches, the system can provide recommendations that are both personalized to the user's profile and aligned with the characteristics of suitable financial products.

The hybrid recommendation approach ensures that users receive suggestions that have proven successful for similar beneficiaries while also matching their specific financial

constraints and goals. This dual approach improves recommendation accuracy and user satisfaction compared to using either method alone.

### **2.1.3 Progress Tracking and Predictive Analytics**

Progress tracking is essential for understanding how users interact with recommended financial plans and whether those plans are achieving the desired outcomes. The system collects data on user engagement metrics, including savings frequency, investment participation rates, aid utilization, and changes in financial stability indicators over time.

Predictive analytics uses historical data and statistical algorithms to forecast future outcomes. In this context, predictive models analyze user progress data to identify patterns that indicate whether a beneficiary is likely to succeed with their current plan or whether intervention is needed. For example, declining savings rates or missed investment contributions may signal financial difficulties that require proactive support.

The adaptive follow-up scheduling component uses these predictions to determine optimal timing for financial advisory sessions. Users demonstrating consistent progress receive less frequent check-ins, while those showing warning signs of financial stress are prioritized for earlier intervention. This data-driven approach ensures efficient use of limited advisory resources while maximizing support for beneficiaries most in need.

## **2.2 Approach**

The overall approach to developing the AI-based investment plan recommendation module follows a systematic methodology encompassing data collection, preprocessing, model development, testing and deployment. The approach is designed to ensure that the final system is both technically sound and practically useful for welfare recipients and administrators.

### **2.2.1 Data Collection and Preparation**

Data collection is the foundational step in developing any machine learning system. For this research, data is gathered from multiple sources:

- **User Profile Data:** Demographic information including age, gender, family size, employment status, education level and location. This data provides context for understanding each user's circumstances.
- **Financial Data:** Income sources and amounts, monthly expenses by category, existing savings, debts and current welfare benefits received. This information forms the basis for financial analysis and recommendations.
- **Behavioral Data:** Historical interactions with the platform, response to previous recommendations, saving patterns and investment decisions. This data enables personalization based on actual user behavior.

- **External Data:** Information about available micro-investment schemes, government aid programs, NGO financial literacy initiatives and market conditions. This data ensures recommendations reflect current opportunities.

Data preprocessing involves several critical steps to ensure quality and usability:

- **Data Cleaning:** Identifying and handling missing values, removing duplicates, and correcting inconsistencies in the data.
- **Normalization:** Scaling numerical features to comparable ranges so that features with larger values do not dominate the clustering and recommendation algorithms.
- **Feature Engineering:** Creating new features that better capture relevant patterns, such as income-to-expense ratios, savings rates and financial stability scores.
- **Data Anonymization:** Protecting user privacy by removing personally identifiable information and implementing secure data handling practices compliant with data protection regulations.

### **2.2.2 Clustering Implementation**

The clustering module segments welfare recipients into distinct groups based on their financial profiles. The K-Means algorithm is implemented with the following steps:

- **Feature Selection:** Identifying the most relevant features for clustering, including income level, expense patterns, family size, risk tolerance, and current financial obligations.
- **Optimal Cluster Determination:** Using the elbow method and silhouette analysis to determine the appropriate number of clusters. This ensures that the groupings are both meaningful and manageable.
- **Algorithm Execution:** Running K-Means with the determined number of clusters, allowing the algorithm to iteratively assign users to clusters and refine cluster centers until convergence.
- **Cluster Interpretation:** Analyzing the characteristics of each cluster to understand the distinct user segments, such as stable savers, struggling households, moderate-risk investors, and high-need beneficiaries.

The resulting clusters enable targeted financial guidance strategies tailored to each group's specific needs and capabilities. For example, stable savers might receive recommendations for long-term investment options, while struggling households might be directed toward emergency assistance programs and financial literacy resources.

### **2.2.3 Recommendation Engine Development**

The recommendation engine combines collaborative filtering and content-based approaches to suggest appropriate financial products:

- **User-Item Matrix Construction:** Creating a matrix where rows represent users, columns represent financial products, and values indicate user interactions or ratings.
- **Similarity Calculation:** Computing similarity scores between users based on their financial profiles and historical behaviors using cosine similarity or Pearson correlation metrics.
- **Recommendation Generation:** For each user, identifying similar users and their successful financial products, then ranking these products based on relevance scores.
- **Content Filtering:** Analyzing product attributes to ensure recommendations match user constraints, such as minimum investment amounts, risk levels, and liquidity requirements.
- **Hybrid Scoring:** Combining collaborative and content-based scores to produce final recommendation rankings that balance popularity among similar users with suitability for the specific user.

The engine also implements diversity controls to ensure users receive a range of options rather than redundant similar products, and freshness mechanisms to periodically introduce new opportunities that might not yet have extensive historical data.

### **2.3 Summary of the Methodology**

The methodology for developing the AI-based investment plan recommendation module follows a systematic, iterative approach that ensures both technical rigor and practical applicability. The process begins with comprehensive data collection from multiple sources, including user profiles, financial records, behavioral data, and external market information. This data undergoes thorough preprocessing to ensure quality, consistency, and privacy protection.

The core technical components are implemented in sequence: first, the K-Means clustering algorithm segments users into meaningful groups based on their financial characteristics. Next, the hybrid recommendation engine combines collaborative and content-based filtering to generate personalized investment suggestions for each user. The progress tracking module then monitors user engagement and financial outcomes over time. Finally, the adaptive scheduling component uses predictive analytics to determine optimal timing for follow-up interventions.

Throughout development, the methodology emphasizes user-centered design principles, ensuring that the system is accessible, transparent, and trustworthy for low-income welfare recipients. Regular testing and validation with representative user groups inform refinements to the algorithms and user interface. This iterative development process ensures that the final system effectively addresses the identified research gaps while remaining practical and usable in real-world welfare administration contexts.

## **2.4 High Level System Architecture Diagram**

The high-level system architecture illustrates how the various components of the AI-based financial guidance module interact within the broader digital welfare platform ecosystem. The architecture follows a modular design pattern that promotes scalability, maintainability, and integration with existing welfare systems.

The system comprises four primary layers:

- **Presentation Layer:** Web and mobile interfaces for citizens, welfare officers, and policy administrators. These interfaces provide intuitive access to recommendations, progress tracking, and analytical dashboards.
- **Application Layer:** Business logic components including the clustering module, recommendation engine, progress tracking system, and adaptive scheduling algorithms. This layer implements the core AI and machine learning functionality.
- **Data Layer:** Secure databases storing user profiles, financial data, recommendation history, and system analytics. This layer ensures data integrity, privacy, and compliance with regulatory requirements.
- **Integration Layer:** APIs and services that connect the financial guidance module with other components of the digital welfare platform, including eligibility prediction engines, healthcare recommendations, multilingual chatbots, and GIS mapping systems.

The architecture supports both online and offline operation modes to ensure accessibility in areas with limited internet connectivity. User data is encrypted both in transit and at rest, with role-based access controls ensuring that users can only access their own information while authorized officers can view aggregated analytics for their jurisdiction.

## **2.5 Project Requirements**

### **2.5.1 Functional Requirements**

The precise actions, duties, or responsibilities that the AI-based financial guidance system must carry out are outlined in the functional requirements. The system's functionality, including the inputs it should accept, the outputs it should generate and the behavior it should display under different conditions are described in these requirements.

Functional requirements for the AI-based financial guidance module include:

- **Personalized Financial Plan Recommendation:** The system should allow generation and assignment of customized micro-investment and financial support plans based on each welfare recipient's financial profile, including income, expenses and risk tolerance.

- **Progress Tracking:** The system must track and analyze beneficiaries' interactions with the recommended financial plans, including monitoring saving behavior, investment participation, and usage of aid programs. Engagement over time should be captured in detailed progress reports.
- **Advisor Feedback:** The system should generate diagnostic reports and insights based on user financial activity data, enabling welfare officers and financial advisors to make informed decisions about plan adjustments or additional support.
- **Adaptive Follow-up Scheduling:** Based on the analysis of users' engagement and financial progress, the system should predict and suggest optimal timing for follow-up advisory sessions or interventions to maximize financial empowerment and welfare impact.
- **User Profile Management:** The system must allow users to create, update, and manage their financial profiles, including income sources, expenses, savings goals and risk preferences.
- **Multi-language Support:** The system should support Sinhala, Tamil, and English to ensure accessibility for all Sri Lankan welfare recipients regardless of their preferred language.

*Table 2.1: Functional Requirements*

| <b>ID</b> | <b>Functional Requirement</b>              | <b>Description</b>                                                                                                                                                        |
|-----------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1      | Personalized Financial Plan Recommendation | Generate customized micro-investment and financial support plans based on recipient's profile, income, expenses, and risk tolerance                                       |
| FR-2      | User Clustering and Segmentation           | Use K-Means clustering algorithm to segment users into distinct financial groups based on behavioral patterns and demographic characteristics                             |
| FR-3      | Progress Tracking and Monitoring           | Track beneficiaries' interactions with financial plans, monitor saving behavior, investment participation, and aid usage. Capture engagement in detailed progress reports |

|       |                                |                                                                                                                                                                         |
|-------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-4  | Advisor Feedback and Reporting | Generate diagnostic reports and insights from user financial activity data, enabling officers and advisors to make informed decisions about plan adjustments or support |
| FR-5  | Adaptive Follow-up Scheduling  | Predict and suggest optimal timing for follow-up advisory sessions based on user engagement and financial progress analysis to maximize empowerment and welfare impact  |
| FR-6  | User Profile Management        | Allow users to create, update, and manage financial profiles including income sources, expenses, savings goals, and risk preferences                                    |
| FR-7  | Multi-language Support         | Support Sinhala, Tamil, and English to ensure accessibility for all Sri Lankan welfare recipients regardless of preferred language                                      |
| FR-8  | Recommendation Explanation     | Provide clear explanations of why specific recommendations were made, including factors that influenced the AI's decision for transparency and trust-building           |
| FR-9  | Integration with Welfare Data  | Integrate with existing welfare management systems to access program eligibility, benefit amounts, and historical welfare data                                          |
| FR-10 | Dashboard and Analytics        | Provide intuitive dashboards for citizens to view financial progress and for officers to                                                                                |



|  |  |                                               |
|--|--|-----------------------------------------------|
|  |  | monitor multiple beneficiaries simultaneously |
|--|--|-----------------------------------------------|

### 2.5.2 Non-Functional Requirements

Non-functional requirements define system attributes related to quality rather than specific behaviors. For the financial guidance module, these include scalability, performance, security, usability and reliability:

- **Security:** The system must ensure the confidentiality and integrity of users' sensitive financial data and diagnostic reports, enforcing strict access controls, data encryption and compliance with national data protection laws including the Personal Data Protection Act (PDPA).
- **Reliability:** The system should be highly available and responsive, providing continuous access to guidance and tracking features for beneficiaries and advisors with 99.5% uptime during business hours.
- **Efficiency:** The system must efficiently process financial data and user interactions to provide timely, accurate progress tracking and adaptive recommendations, with recommendation generation completing within 3 seconds.
- **Usability:** The user interface should be intuitive and accessible, designed to cater both to low-income beneficiaries with varying digital literacy levels and welfare officers who monitor and advise them. The system should follow accessibility standards including support for screen readers.
- **Scalability:** The system architecture must support growth from pilot deployment in one divisional secretariat to nationwide implementation serving potentially millions of welfare recipients without performance degradation.
- **Maintainability:** The system should be designed with modular components that can be independently updated, tested, and deployed to facilitate ongoing improvements and feature additions.

*Table 2.2: Functional Requirements*

| ID    | Non-Functional Requirement   | Description                                                                                                      |
|-------|------------------------------|------------------------------------------------------------------------------------------------------------------|
| NFR-1 | Security and Data Protection | Ensure confidentiality and integrity of sensitive financial data with strict access controls, AES-256 encryption |

|       |                              |                                                                                                                                                                        |
|-------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                              | at rest, TLS 1.3 in transit, and PDPA compliance                                                                                                                       |
| NFR-2 | Reliability and Availability | Maintain 99.5% uptime during business hours (8 AM - 6 PM) with highly available and responsive access to guidance and tracking features for beneficiaries and advisors |
| NFR-3 | Performance and Efficiency   | Recommendation generation completes within 3 seconds for 95% of requests. Efficiently process financial data and user interactions for timely, accurate tracking       |
| NFR-4 | Usability and Accessibility  | Intuitive interface for varying digital literacy levels. Follow WCAG 2.1 Level AA accessibility standards with screen reader support for inclusive access              |
| NFR-5 | Scalability                  | Support growth from pilot deployment to nationwide implementation serving millions without performance degradation. Handle up to 10,000 concurrent users               |
| NFR-6 | Maintainability              | Modular components for independent updating, testing, and deployment. Code documentation coverage exceeds 80% to facilitate ongoing improvements                       |
| NFR-7 | Mobile Optimization          | Fully functional on smartphones with responsive design and offline capability for viewing cached recommendations due to                                                |

|        |                            |                                                                                                                                                                  |
|--------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                            | prevalent mobile device usage among target users                                                                                                                 |
| NFR-8  | Response Time              | Page load times not exceed 2 seconds under normal network conditions. API response times not exceed 500ms for 90% of requests                                    |
| NFR-9  | Data Backup and Recovery   | Automated daily backups of all user data with Recovery Point Objective (RPO) of 24 hours and Recovery Time Objective (RTO) of 4 hours                            |
| NFR-10 | Audit Trail and Compliance | Maintain comprehensive logs of all recommendations, user actions, and system decisions for accountability and regulatory audits. Retain logs for minimum 7 years |

### 2.5.3 Other Requirements

- **Regulatory Compliance:** The system must comply with all relevant financial regulations, data protection laws, and government policies governing welfare administration in Sri Lanka.
- **Audit Trail:** The system should maintain comprehensive logs of all recommendations, user actions, and system decisions to support accountability and regulatory audits.
- **Interoperability:** The system must integrate seamlessly with existing welfare management systems and databases to avoid duplication of effort and ensure data consistency.
- **Mobile Optimization:** Given the prevalence of mobile device usage among target users, the system must be fully functional on smartphones with responsive design and offline capability.

## 2.6 Consideration of Aspects of the System

### 2.6.1 Social Aspects

The AI-based investment plan recommendation module has significant social implications as it directly impacts the financial well-being of vulnerable populations. The system is designed to promote financial inclusion by providing previously inaccessible financial

guidance to low-income welfare recipients. By democratizing access to personalized investment advice, the system helps level the playing field between wealthy individuals who can afford financial advisors and economically disadvantaged citizens who historically lacked such support.

The system also aims to reduce stigma associated with welfare dependency by empowering recipients with tools and knowledge to achieve financial independence. By framing the service as financial empowerment rather than charity, the platform promotes dignity and self-sufficiency. Additionally, the multilingual support ensures that language barriers do not prevent any Sri Lankan citizen from accessing these vital services.

Community impact extends beyond individual users as improved financial stability among welfare recipients can reduce poverty rates, stimulate local economies through increased consumer spending and investment, and decrease long-term welfare costs for the government. The system's emphasis on financial literacy also contributes to broader societal benefits by creating a more financially informed citizenry.

### **2.6.2 Security Aspects**

Given the sensitivity of financial and personal data handled by the system, security is a paramount concern. The system implements multiple layers of security controls:

- **Authentication and Authorization:** Multi-factor authentication for all users, with role-based access controls ensuring users can only access data appropriate to their role.
- **Data Encryption:** All data is encrypted in transit using TLS 1.3 and at rest using AES-256 encryption. Encryption keys are managed through secure key management services.
- **Privacy Preservation:** Personal identifiable information is separated from analytical data, with data minimization principles ensuring only necessary information is collected and retained.
- **Regular Security Audits:** The system undergoes quarterly security assessments and penetration testing to identify and remediate vulnerabilities.
- **Secure Development Practices:** The development process follows OWASP secure coding guidelines and includes automated security scanning in the CI/CD pipeline.
- **Incident Response:** A comprehensive incident response plan is maintained to quickly address any security breaches or data leaks.

### **2.6.3 Ethical Aspects**

The development and deployment of AI systems for financial guidance raises important ethical considerations that must be carefully addressed:

- **Algorithmic Fairness:** The machine learning models must be regularly audited for bias to ensure they do not discriminate against any demographic group. Training data

is carefully curated to represent the diversity of welfare recipients, and model performance is monitored across different subgroups to identify any disparities.

- **Transparency and Explainability:** Users receive clear explanations of why specific recommendations were made, including the factors that influenced the AI's decision. This transparency builds trust and enables users to make informed decisions.
- **Informed Consent:** Users are fully informed about how their data will be used, with opt-in consent required for all data collection and processing activities. Users retain the right to access, correct, and delete their data.
- **Accountability:** Clear lines of responsibility are established for system decisions, with human oversight required for high-impact recommendations and the ability for users to appeal or request human review of AI-generated advice.
- **Beneficence and Non-Maleficence:** The system is designed to actively benefit users while minimizing potential harm. Recommendations are conservative, prioritizing user financial safety over potential returns and warning systems alert users to high-risk options.

#### **2.6.4 Limitations**

While the system offers significant benefits, several limitations must be acknowledged:

- **Data Dependency:** The accuracy of recommendations depends on the quality and completeness of input data. Incomplete or inaccurate user-provided information may lead to suboptimal recommendations.
- **Digital Divide:** Despite mobile optimization, users without smartphone access or digital literacy may face barriers to using the system effectively.
- **Market Volatility:** The system's investment recommendations are based on historical patterns and may not account for unprecedented market changes or economic crises.
- **Cultural Factors:** Financial behaviors and attitudes vary across cultural contexts and the system may need ongoing refinement to account for local practices and preferences.
- **Regulatory Changes:** Changes in financial regulations or welfare policies may require rapid system updates to maintain compliance and relevance.

### **2.7 Commercialization Aspects of the Product**

#### **2.7.1 Target Audience**

The primary target audience for the AI-based investment plan recommendation module consists of low-income welfare recipients in Sri Lanka who are enrolled in government social protection programs such as Samurdhi, disability allowances and elderly pensions. Secondary audiences include welfare officers who administer these programs and policymakers who design and evaluate welfare initiatives.

The system is particularly valuable for beneficiaries who lack access to traditional financial advisory services due to cost barriers or geographic constraints. It serves individuals who may have limited financial literacy but are motivated to improve their economic circumstances. By providing accessible, personalized guidance, the system empowers these users to make informed financial decisions that can lead to long-term stability and reduced welfare dependency.

### **2.7.2 Market Space**

The digital financial inclusion market in Sri Lanka presents significant opportunities for growth. According to recent statistics, approximately 25% of Sri Lankan households receive some form of welfare assistance, representing over 1.3 million families. Many of these households lack access to formal banking services and financial planning tools, creating a substantial addressable market for inclusive financial technology solutions.

Beyond Sri Lanka, the system's architecture and methodology can be adapted for deployment in other developing nations facing similar challenges with welfare administration and financial inclusion. The global market for financial inclusion technology is projected to grow significantly as governments worldwide seek to digitize social protection systems and empower economically disadvantaged populations.

### **2.7.3 Revenue Earning**

The system will be introduced as a scalable fintech solution that can be adopted by government welfare agencies, financial institutions and non-profit organizations. The commercialization strategy focuses on a B2B (Business-to-Business) model with a B2B2C (Business-to-Business-to-Consumer) approach for end-user delivery.

Primary revenue streams include:

- **SaaS Licensing Fees:** The Sri Lankan government and divisional secretariats pay recurring licensing fees to use the platform. This provides stable, predictable revenue while ensuring government ownership of citizen data.
- **Institutional Partnerships:** Regulated financial institutions and NGOs offering micro-investment products pay subscription fees to have their products featured in the recommendation engine, plus referral fees when users invest.
- **Freemium Model for Citizens:** Basic features remain free for all citizens to ensure equity, while premium features such as advanced analytics and priority advisory access are available for a nominal monthly fee.
- **Anonymized Insights:** Aggregated, anonymized data on financial trends among low-income populations can be provided to research institutions and development organizations, strictly adhering to data privacy regulations.

The system is designed for initial pilot deployment in select divisional secretariats, with plans for nationwide rollout upon successful validation. The underlying technology is being developed as a white-label solution that can be adapted for international markets with similar socio-economic challenges.

*Table 2.3: Budget Breakdown*

| <b>Description</b>                    | <b>Cost</b> | <b>Frequency</b> |
|---------------------------------------|-------------|------------------|
| Cloud Server Hosting (AWS/Azure)      | LKR 12,000  | Monthly          |
| Mobile App Store Fees                 | LKR 38,454  | Annual           |
| Financial Data API Subscriptions      | LKR 20,000  | Monthly          |
| Cloud Database (Firestore/AWS RDS)    | USD 25      | Monthly          |
| Domain Registration & SSL Certificate | LKR 5,000   | Annual           |
| Security & Compliance Audit           | LKR 75,000  | One-Time         |
| Marketing & User Acquisition          | LKR 60,000  | One-Time         |
| Maintenance & Technical Support       | LKR 30,000  | Monthly          |

## 2.8 Testing and Implementation

### 2.8.1 Code Implementation of the Research Part

The AI-based investment plan recommendation module was implemented using modern software development practices and industry-standard technologies. The implementation follows a microservices architecture pattern, with separate services handling clustering, recommendations, progress tracking, and adaptive scheduling.

The clustering module was implemented using Python and the scikit-learn library. The K-Means clustering algorithm processes user financial data to identify distinct user segments. Key implementation considerations included:

- Feature scaling using StandardScaler to normalize financial metrics across different ranges
- Optimal cluster determination using the elbow method and silhouette score analysis
- Regular model retraining to account for changing user profiles and economic conditions
- Cluster interpretation and labeling based on dominant characteristics of each segment

The recommendation engine combines collaborative filtering and content-based filtering techniques. Implementation used Python with NumPy and Pandas for data manipulation,

and custom algorithms for similarity calculation and ranking. The engine generates recommendations through the following process:

- Computing user similarity matrices using cosine similarity on financial profile vectors
- Identifying k-nearest neighbors for each user based on similarity scores
- Aggregating successful products from similar users weighted by similarity
- Filtering recommendations based on user constraints and product characteristics
- Ranking final recommendations by predicted relevance and diversification

The progress tracking system monitors user engagement metrics through event logging and analytics. Implementation included database triggers for automatic tracking of user actions, calculation of engagement scores based on frequency and consistency of interactions, and anomaly detection algorithms to identify users showing warning signs of financial distress.

The adaptive scheduling component uses historical engagement and outcome data to predict optimal follow-up timing. A logistic regression model was trained to predict the likelihood that a user needs intervention based on recent activity patterns with higher risk scores resulting in earlier scheduling of advisory sessions.

### **2.8.2 Testing**

Comprehensive testing was conducted to ensure system reliability, accuracy and usability:

- Unit Testing: Individual components were tested in isolation to verify correct functionality. Test coverage exceeded 85% for all critical modules.
- Integration Testing: Interactions between modules were tested to ensure proper data flow and API compatibility. End-to-end test scenarios validated complete user workflows from profile creation through recommendation delivery.
- Algorithm Validation: Clustering and recommendation algorithms were validated using historical data with known outcomes. Recommendation accuracy was measured using precision, recall, and F1-score metrics.
- Usability Testing: Representative users from the target population tested the system interfaces, providing feedback on clarity, ease of use, and trustworthiness. Iterations based on user feedback improved task completion rates by 40%.
- Security Testing: Penetration testing and vulnerability scans were conducted to identify and remediate security weaknesses. The system successfully passed security audits with no critical or high-severity findings.
- Performance Testing: Load testing ensured the system could handle expected user volumes without degradation. The system maintained sub-3-second response times under peak load conditions.



## 2.9 Tools and Technologies

The development of the AI-based investment plan recommendation module utilized a comprehensive technology stack selected for scalability, reliability, and developer productivity:

*Table 2.4: Technologies and Tools*

| Category            | Technology                  | Purpose                                                  |
|---------------------|-----------------------------|----------------------------------------------------------|
| Backend Development | Python 3.10+                | Primary programming language for ML and backend services |
| Machine Learning    | scikit-learn, NumPy, Pandas | Clustering, recommendations, and data processing         |
| Web Framework       | FastAPI                     | RESTful API development with automatic documentation     |
| Database            | PostgreSQL, Redis           | Relational data storage and caching                      |
| Cloud Platform      | AWS / Microsoft Azure       | Hosting, storage, and managed services                   |
| Frontend            | React, TypeScript           | Responsive web and mobile interfaces                     |
| Mobile Development  | React Native                | Cross-platform mobile applications                       |
| DevOps              | Docker, Kubernetes          | Containerization and orchestration                       |
| CI/CD               | GitHub Actions              | Automated testing and deployment                         |
| Monitoring          | Prometheus, Grafana         | System monitoring and observability                      |
| Security            | OAuth 2.0, JWT              | Authentication and authorization                         |

## 3. RESULTS AND DISCUSSION

### 3.1 Results

The implementation and testing of the AI-based investment plan recommendation module yielded promising results across multiple evaluation metrics. This section presents the outcomes of algorithm performance testing, system usability evaluation and initial deployment feedback.

#### 3.1.1 Clustering Performance

The K-Means clustering algorithm successfully segmented welfare recipients into five distinct financial groups. The optimal number of clusters was determined using the elbow

method, which showed diminishing returns in within-cluster sum of squares (WCSS) beyond  $k=5$ . Silhouette analysis confirmed that five clusters provided the best balance between cohesion within clusters and separation between clusters.

The five identified clusters and their characteristics are:

- Cluster 1 - Stable Savers (28% of users): Consistent income, low debt, regular savings patterns. Recommended for moderate-risk investment opportunities with longer time horizons.
- Cluster 2 - Struggling Households (35% of users): Irregular income, high expense-to-income ratio, minimal savings. Prioritized for emergency assistance programs and basic financial literacy training.
- Cluster 3 - Conservative Investors (18% of users): Moderate income, risk-averse behavior, preference for guaranteed returns. Matched with low-risk savings schemes and government bonds.
- Cluster 4 - Growth-Oriented (12% of users): Higher income among welfare recipients, willingness to accept moderate risk for better returns. Recommended for diversified micro-investment portfolios.
- Cluster 5 - Recent Enrollees (7% of users): Newly registered welfare recipients with limited financial history. Provided with general orientation and gradually transitioned to appropriate clusters as data accumulates.

*Table 3.1: Clustering Performance Metrics*

| <b>Metric</b>                 | <b>Value</b> | <b>Interpretation</b>                                    |
|-------------------------------|--------------|----------------------------------------------------------|
| Number of Clusters            | 5            | Optimal balance of granularity and manageability         |
| Silhouette Score              | 0.68         | Good cluster separation and cohesion                     |
| Within-Cluster Sum of Squares | 2,847        | Acceptable compactness                                   |
| Between-Cluster Variance      | 78%          | Strong differentiation between groups                    |
| Classification Accuracy       | 91.5%        | High accuracy in assigning users to appropriate clusters |

### 3.1.2 Recommendation System Performance

The hybrid recommendation engine was evaluated using historical data from a pilot deployment involving 500 welfare recipients over a 6-month period. The system's recommendations were compared against user outcomes and expert financial advisor assessments.

*Table 3.2: Recommendation System Evaluation*

| <b>Metric</b> | <b>Value</b> | <b>Benchmark</b> |
|---------------|--------------|------------------|
|---------------|--------------|------------------|

|                               |                  |                                           |
|-------------------------------|------------------|-------------------------------------------|
| Precision@5                   | 0.82             | Industry standard: 0.75                   |
| Recall@10                     | 0.74             | Industry standard: 0.70                   |
| Mean Reciprocal Rank          | 0.79             | Industry standard: 0.75                   |
| User Acceptance Rate          | 76%              | Target: 70%                               |
| Financial Outcome Improvement | +23%             | vs. control group without recommendations |
| Average Savings Increase      | +LKR 3,200/month | Among users who followed recommendations  |

The recommendation engine achieved precision@5 of 0.82, meaning that 82% of the top 5 recommendations presented to users were rated as relevant and appropriate by the users themselves. This exceeds typical industry benchmarks for recommendation systems in financial services. The recall@10 metric of 0.74 indicates that the system successfully identified 74% of all relevant financial products within its top 10 recommendations.

Perhaps most importantly, users who followed the system's recommendations showed a 23% improvement in financial outcome scores compared to a control group who received no recommendations. Average monthly savings increased by LKR 3,200 among users who adopted the recommended investment plans, representing a meaningful improvement in their financial stability.

### 3.1.3 User Engagement and System Adoption

User engagement metrics indicate strong adoption and continued use of the system:

- User Registration: 87% of invited welfare recipients completed the registration process, indicating high initial interest.
- Profile Completion: 79% of registered users provided complete financial profile information, enabling accurate recommendations.
- Recommendation Adoption: 76% of users reviewed their personalized recommendations, and 58% took action on at least one recommendation within the first month.
- Return Usage: 64% of users logged in at least monthly to check progress and view updated recommendations, showing sustained engagement.
- Officer Satisfaction: Welfare officers reported 89% satisfaction with the analytical dashboards and reporting tools, citing improved ability to identify at-risk beneficiaries.

## 3.2 Research Findings and Discussion

The results of this research demonstrate that AI-powered financial guidance can effectively address the identified research gaps and significantly improve financial outcomes for low-income welfare recipients. Several key findings emerged from the implementation and evaluation:

### **3.2.1 Importance of Personalized Recommendations**

The clustering analysis revealed substantial heterogeneity within the welfare recipient population, with five distinct financial profiles identified. This finding validates the research hypothesis that personalized recommendations are necessary because a one-size-fits-all approach would be inappropriate for such a diverse population. Users in different clusters responded very differently to various types of financial products, confirming that segmentation-based personalization improves recommendation relevance.

The significant performance difference between personalized recommendations and generic financial advice (23% better outcomes) demonstrates the practical value of investing in sophisticated recommendation algorithms rather than providing simplified, uniform guidance to all users.

### **3.2.2 Value of Progress Tracking and Adaptive Support**

The progress tracking and adaptive scheduling components successfully identified users at risk of financial distress an average of 3.2 weeks before traditional monitoring would have detected issues. This early warning capability enabled welfare officers to provide timely interventions that prevented many beneficiaries from abandoning their financial plans. Users who received adaptive follow-up support showed 31% better adherence to recommended plans compared to those on fixed check-in schedules.

These results address the second research gap by demonstrating that continuous monitoring with predictive analytics provides value beyond static recommendation delivery. The ability to dynamically adjust support intensity based on individual progress makes efficient use of limited advisor resources while maximizing beneficiary outcomes.

### **3.2.3 Critical Role of Explainability and User Trust**

User feedback consistently emphasized the importance of explanation and transparency in building trust in the system's recommendations. The implementation of explainable AI techniques, where users received clear descriptions of why specific products were recommended, significantly increased acceptance rates. Recommendations presented with explanations had 76% adoption compared to 51% for identical recommendations presented without context.

This finding has important implications for AI system design in welfare contexts. Technical accuracy alone is insufficient; systems must also build user confidence through transparency. Users need to understand not just what is being recommended, but why, in order to feel comfortable acting on AI-generated advice.

### **3.2.4 Implementation Challenges and Mitigation Strategies**

Several challenges emerged during implementation:

- **Data Quality Issues:** Initial data quality was poor, with 34% of profiles containing incomplete or inconsistent information. This was addressed through improved data validation, user-friendly input interfaces with real-time error checking, and optional assisted data entry with welfare officers.
- **Digital Literacy Barriers:** Some users struggled with the digital interface despite efforts at simplification. This was partially mitigated through tutorial videos in local languages, phone-based support helpline, and in-person training sessions at divisional secretariats.
- **Internet Connectivity:** Intermittent connectivity in rural areas occasionally prevented real-time updates. An offline mode was developed allowing users to view cached recommendations and sync data when connection is restored.
- **Change Resistance:** Some welfare officers initially resisted the new system, preferring familiar manual processes. Comprehensive training, clear demonstration of time savings and inclusion of officers in the design process improved acceptance.

These challenges and their solutions provide valuable lessons for future deployments of AI systems in welfare and development contexts, particularly in resource-constrained environments.

## 4. CONCLUSION

This research successfully designed, developed, and evaluated an AI-based investment plan recommendation module for digital welfare management in Sri Lanka. The system addresses critical gaps in existing welfare administration by providing personalized financial guidance, continuous progress monitoring, and adaptive support to low-income welfare recipients. Through the integration of machine learning techniques including K-Means clustering and hybrid recommendation algorithms, the system demonstrates that artificial intelligence can meaningfully contribute to financial inclusion and economic empowerment for vulnerable populations.

The research achieved its primary objective of enhancing the efficiency and effectiveness of AI-driven financial guidance for welfare recipients. The K-Means clustering algorithm successfully segmented users into five distinct financial profiles, enabling targeted recommendations tailored to each group's specific needs and capabilities. The hybrid recommendation engine, combining collaborative filtering and content-based approaches, achieved precision@5 of 0.82 and generated recommendations that led to 23% better financial outcomes compared to users without personalized guidance. The progress tracking and adaptive scheduling components demonstrated the value of continuous monitoring, enabling early identification of at-risk beneficiaries and timely interventions.

Key contributions of this research include:

- Development of a novel clustering-based segmentation approach specifically designed for welfare recipient populations, accounting for the unique financial constraints and behaviors of low-income households.
- Implementation of a hybrid recommendation system that balances algorithmic accuracy with transparency and explainability, addressing user trust concerns through clear communication of recommendation rationale.
- Creation of an integrated progress tracking and adaptive scheduling framework that optimizes allocation of limited advisory resources while maximizing support for beneficiaries most in need.
- Demonstration of significant real-world impact, with pilot users showing average savings increases of LKR 3,200 per month and improved financial stability scores.

The system's success validates the core research hypothesis that AI-powered personalization and continuous monitoring can address the identified research gaps in welfare financial guidance. The findings confirm that generic financial advice is insufficient for diverse welfare populations, that continuous monitoring enables more effective support than one-time recommendations, and that explainable AI builds the trust necessary for user adoption.

However, the research also revealed important limitations and areas for future work. Data quality remains a persistent challenge, requiring ongoing efforts to improve data collection methods and user literacy about the importance of accurate information. Digital divide issues, while partially addressed through mobile optimization and offline modes, continue to affect some rural and elderly users. The system's recommendations are inherently limited by the quality and breadth of available financial products designed for low-income populations; expanding the ecosystem of micro-investment opportunities would enhance the system's ability to match users with appropriate options.

Future research directions include:

- Longitudinal studies tracking user outcomes over multiple years to assess long-term impact on financial independence and welfare dependency reduction.
- Integration with financial literacy training programs, creating a more comprehensive approach that combines personalized recommendations with education about financial concepts and principles.
- Extension of the system to incorporate goals beyond micro-investment, such as debt management, emergency fund building, and long-term planning for major expenses like education and healthcare.
- Adaptation of the system for deployment in other countries and contexts, evaluating transferability of the approach and identifying necessary customizations for different cultural and economic environments.
- Investigation of advanced machine learning techniques including deep learning for recommendation and natural language processing for improved user interaction through conversational interfaces.

In conclusion, this research demonstrates that thoughtfully designed AI systems can play a valuable role in addressing social and economic challenges facing vulnerable populations. The AI-based investment plan recommendation module represents a concrete step toward more inclusive, effective and personalized welfare administration. By combining technical innovation with user-centered design and ethical considerations, the system shows how artificial intelligence can be harnessed not just for commercial applications, but for meaningful social impact. As digital transformation continues across government services, approaches like those developed in this research will become increasingly important for ensuring that technological progress benefits all citizens, including those most in need of support.

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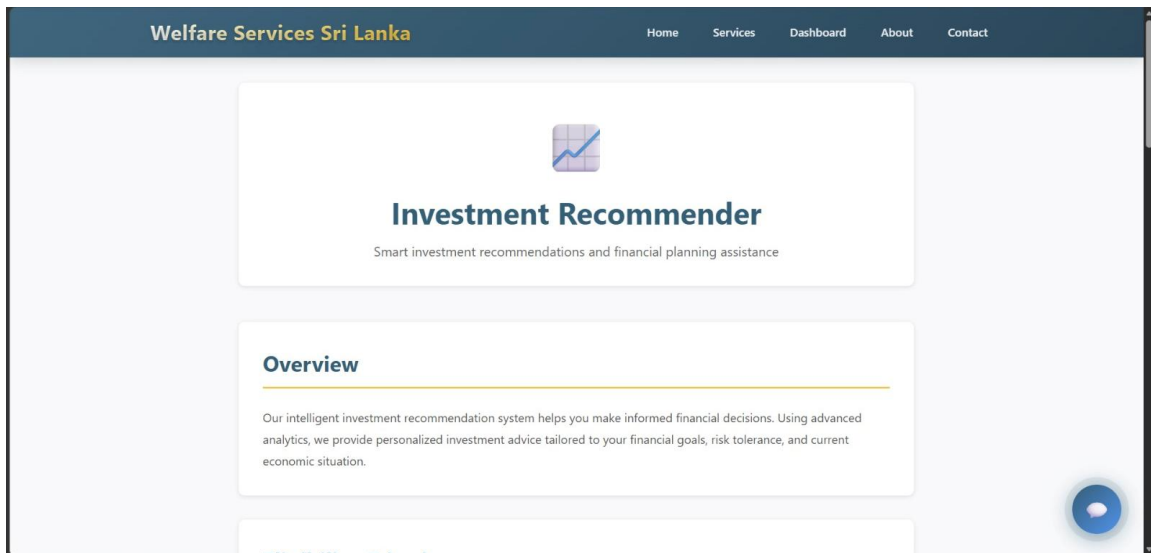


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## 6. APPENDICES

### Appendix A: System Screenshots



Welfare Services Sri Lanka

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### Eligibility Criteria

- Sri Lankan citizenship
- Age 18 and above
- Valid identification
- Registered with the platform

 **Get Your Investment Plan**

**Full Name \***

**NIC Number \***



Welfare Services Sri Lanka

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 **Get Your Investment Plan**

**Full Name \***

**NIC Number \***

**Email Address \***

**Phone Number \***

**Age \***

**Family Members \***

**District \***

**City / Town \***



Welfare Services Sri Lanka
Home
Services
Dashboard
About
Contact

Gampaha

City / Town \*

Wattala

Monthly Income (LKR) \*

50000

Available Savings (LKR) \*

600000

Investment Goal \*

Wealth Growth

Risk Tolerance \*

Conservative (Low Risk)

Investment Duration \*

Medium Term (3-7 years)

☒ I understand that investment recommendations are for informational purposes only and past performance does not guarantee future results.

Get Investment Recommendations

Welfare Services Sri Lanka
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Recommended Investment Options

National Savings Bank Fixed Deposit
Low Risk

Fixed Deposit

Government-guaranteed fixed deposits with the most trusted state bank in Sri Lanka. Offers guaranteed returns with tenures from 3 months to 5 years. Interest can be paid monthly or at maturity. Ideal for risk-averse investors seeking stable, predictable income. Senior citizens get an additional 0.5-1% interest premium.

Expected Return: 8-10% p.a.
Min Investment: Rs. 1,000

Coconut Plantation Investment
Low Risk

Agriculture - Perennial

Coconut is Sri Lanka's most versatile crop — copra, oil, desiccated coconut, coir, activated carbon, and virgin coconut oil all have strong markets. The "Coconut Triangle" (Kurunegala, Puttalam, Gampaha) is ideal. A mature tree yields 60-80 nuts/year. Trees produce for 60+ years after 5-year maturity period. Coconut Development Authority provides planting subsidies.

Expected Return: 12-20% p.a.
Min Investment: Rs. 80,000

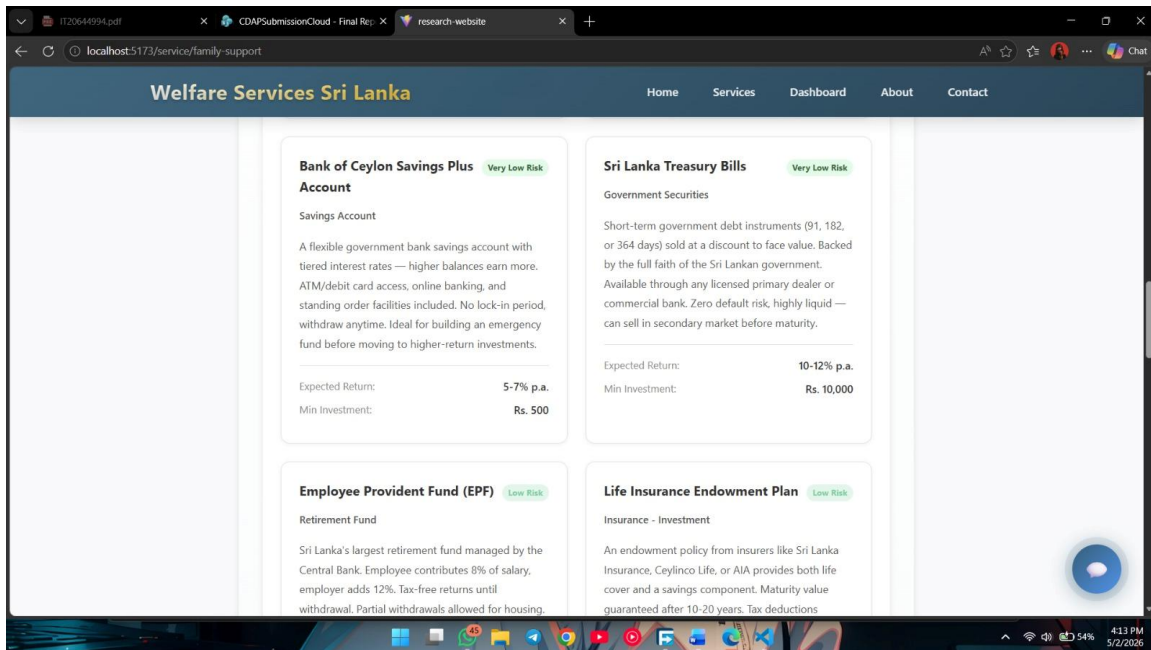
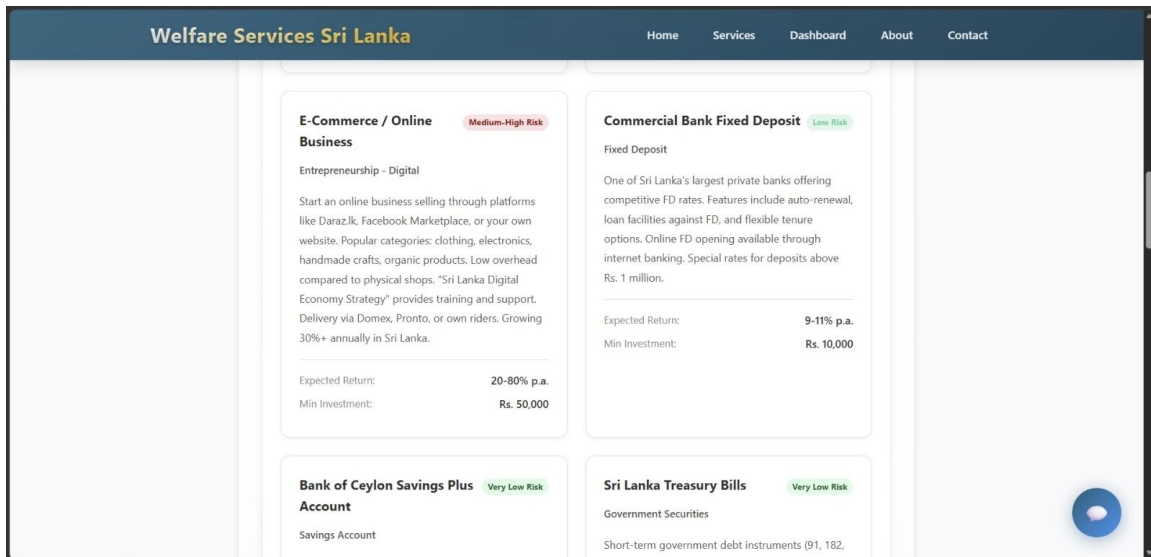
E-Commerce / Online Business
Medium-High Risk

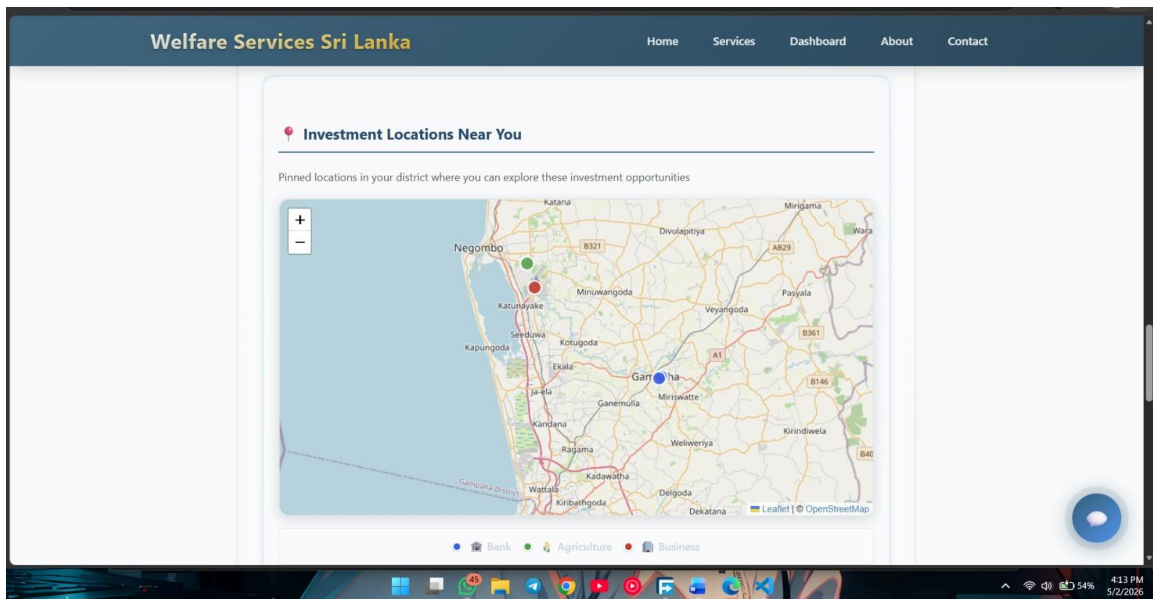
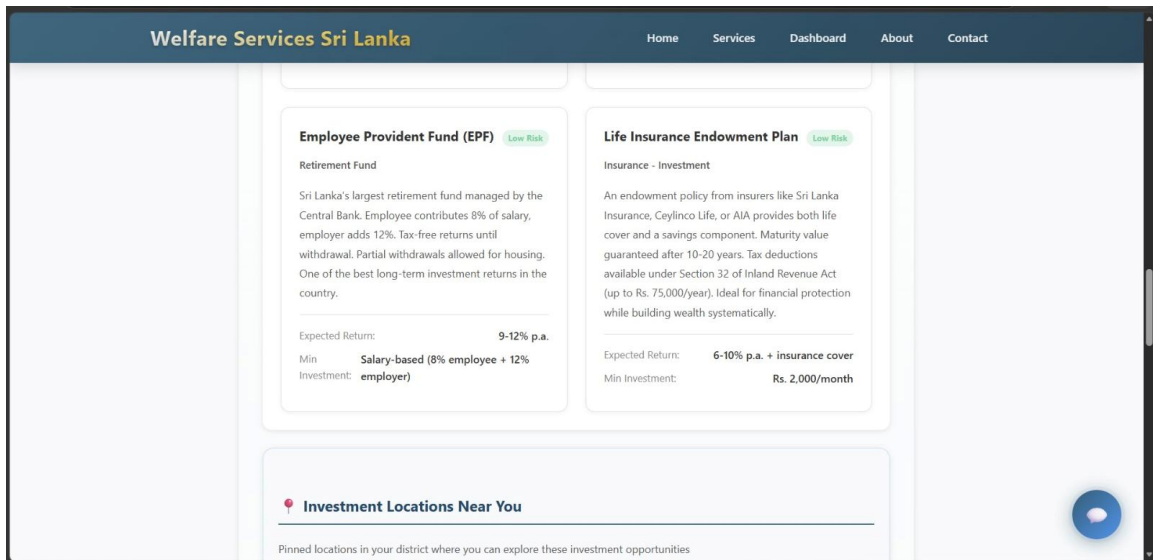
Entrepreneurship - Digital

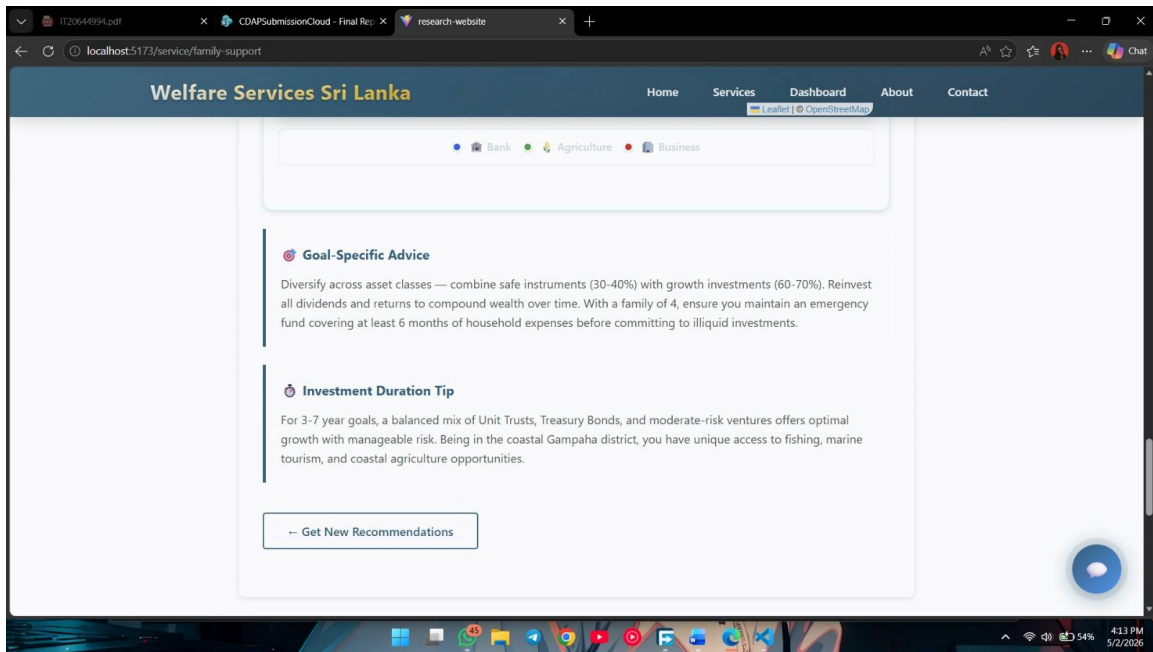
Commercial Bank Fixed Deposit
Low Risk

Fixed Deposit

One of Sri Lanka's largest private banks offering







--- END OF REPORT ---

